CSE113
Foundations of Computer Science

Proof By Contraposition

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Review Exercise

- Are P-> Q and~Q -> ~P equivalent, and why?

Contraposition, Contrapostive

- contraposition = inference of going from a conditional statement into its logically equivalent contrapositive

- P -> Q
  - contrapositive is ~Q -> ~P

- "If it is raining, then I wear my coat”
  - contrapositive is "If I don't wear my coat, then it isn't raining.”

Proof by contraposition

- Proposition: P -> Q
- Proof
  - Proof by contraposition: We want to prove ~Q -> ~P
  - Suppose ~Q
  - ....
  - Therefore ~P
- QED.

Example 1

- Suppose x is an arbitrary integer. Prove: If $7x + 9$ is even, then x is odd.

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Proposition Suppose $x \in \mathbb{Z}$. If $7x+9$ is even, then x is odd.

Proof. (Direct) Suppose $7x+9$ is even.
Thus $7x+9 = 2a$ for some integer a.
Subtracting $6x+9$ from both sides, we get $x = 2a - 6x - 9$.
Thus $x = 2a - 6x - 9 = 2a - 6x - 10 + 1 = 2(a - 3x - 5) + 1$.
Consequently $x = 2b + 1$, where $b = a - 3x - 5 \in \mathbb{Z}$.
Therefore x is odd. ■

Here is a contrapositive proof of the same statement:

Proposition Suppose $x \in \mathbb{Z}$. If $7x+9$ is even, then x is odd.

Proof. (Contrapositive) Suppose x is not odd.
Thus x is even, so $x = 2a$ for some integer a.
Then $7x+9 = 7(2a)+9 = 14a+8+1 = 2(7a+4)+1$.
Therefore $7x+9 = 2b+1$, where b is the integer $7a+4$.
Consequently $7x+9$ is odd.
Therefore $7x+9$ is not even. ■

Example 2

Prove: If $x^2 - 6x + 5$ is even, then x is odd.

Proposition Suppose $x \in \mathbb{Z}$. If $x^2 - 6x + 5$ is even, then x is odd.

Proof. (Contrapositive) Suppose x is not odd.
Thus x is even, so $x = 2a$ for some integer a.
So $x^2 - 6x + 5 = (2a)^2 - 6(2a) + 5 = 4a^2 - 12a + 5 = 4a^2 - 12a + 4 + 1 = 2(2a^2 - 6a + 2) + 1$.
Therefore $x^2 - 6x + 5 = 2b + 1$, where b is the integer $2a^2 - 6a + 2$.
Consequently $x^2 - 6x + 5$ is odd.
Therefore $x^2 - 6x + 5$ is not even. ■

Example 3

Prove: Suppose $x, y \in \mathbb{R}$. If $y^3 + yx^2 \leq x^3 + xy^2$, then $y \leq x$.

Proof. (Contrapositive) Suppose it is not true that $y \leq x$, so $y > x$.
Then $y - x > 0$. Multiply both sides of $y - x > 0$ by the positive value $x^2 + y^2$.

$$\begin{aligned}
(y-x)(x^2+y^2) &> 0(x^2+y^2) \\
yx^2 + y^3 - x^3 - xy^2 &> 0 \\
y^3 + yx^2 &> x^3 + xy^2
\end{aligned}$$

Therefore $y^3 + yx^2 > x^3 + xy^2$, so it is not true that $y^3 + yx^2 \leq x^3 + xy^2$. ■

Example 4

Prove: If $x^2 - 6x + 5$ is even, then x is odd.

Proof. (Contrapositive) Suppose x is not odd.
Thus x is even, so $x = 2a$ for some integer a.
So $x^2 - 6x + 5 = (2a)^2 - 6(2a) + 5 = 4a^2 - 12a + 5 = 4a^2 - 12a + 4 + 1 = 2(2a^2 - 6a + 2) + 1$.
Therefore $x^2 - 6x + 5 = 2b + 1$, where b is the integer $2a^2 - 6a + 2$.
Consequently $x^2 - 6x + 5$ is odd.
Therefore $x^2 - 6x + 5$ is not even. ■

Example 5

Prove: Suppose $x, y \in \mathbb{Z}$. If $5 \nmid xy$, then $5 \nmid x$ and $5 \nmid y$.

Proof. (Contrapositive) Suppose it is not true that $5 \nmid x$ **and** $5 \nmid y$.
By DeMorgan's law, it is not true that $5 \nmid x$ **or** it is not true that $5 \nmid y$.
Therefore $5 \mid x$ or $5 \mid y$. We consider these possibilities separately.
Case 1. Suppose $5 \mid x$. Then $x = 5a$ for some $a \in \mathbb{Z}$.
From this we get $xy = 5(ay)$, and that means $5 \mid xy$.
Case 2. Suppose $5 \mid y$. Then $y = 5a$ for some $a \in \mathbb{Z}$.
From this we get $xy = 5(ax)$, and that means $5 \mid xy$.
The above cases show that $5 \mid xy$, so it is not true that $5 \nmid xy$. ■

Exercises

Warm up

1. Suppose $n \in \mathbb{Z}$. If n^2 is even, then n is even.

2. Suppose $n \in \mathbb{Z}$. If n^2 is odd, then n is odd.

3. Suppose $a, b \in \mathbb{Z}$. If $a^2(b^2 - 2b)$ is odd, then a and b are odd.

4. Suppose $a, b, c \in \mathbb{Z}$. If a does not divide bc, then a does not divide b.

5. Suppose $x \in \mathbb{R}$. If $x^2 + 5x < 0$ then $x < 0$.

6. Suppose $x \in \mathbb{R}$. If $x^3 - x > 0$ then $x > -1$.

9. Suppose $n \in \mathbb{Z}$. If $3 \nmid n^2$, then $3 \nmid n$.

10. Suppose $x, y, z \in \mathbb{Z}$ and $x \neq 0$. If $x \nmid yz$, then $x \nmid y$ and $x \nmid z$.

11. Suppose $x, y \in \mathbb{Z}$. If $x^2(y+3)$ is even, then x is even or y is odd.

n^2 is even $\implies$ n is even

(Same as: Prove: For any integer n, If n^2 is even, then n is even)

Exercise 2: Prove the following

Suppose $x, y \in \mathbb{Z}$. If $x^2(y+3)$ is even, then x is even or y is odd.

Exercise 3: Prove the following

Suppose $x \in \mathbb{R}$. If $x^5 - 4x^4 + 3x^3 - x^2 + 3x - 4 \geq 0$, then $x \geq 0$.

Exercise 4: Prove the following

Suppose $a \in \mathbb{Z}$. If a^2 is not divisible by 4, then a is odd.

Exercise 5: Prove the following

Suppose $x \in \mathbb{R}$. If $x^5 + 7x^3 + 5x \geq x^4 + x^2 + 8$, then $x \geq 0$.